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Improving medical students' learning absorption in a knowledge graph based blended learning course

Bugang Li[1†], Wenhui Hu[1,2†], Anyan Duan[1], Qun Li[1,3], Lin Li[1], Chunyun Li[1], Chunyan Weng[4], Zekang Peng[1], Wei Chen[1*] and Yufei Li[1*]

Abstract

Background Learning engagement refers to the time and effort students invest in effective learning practices, and it serves as a crucial indicator of their learning abilities as well as a significant proximal factor in predicting their academic performance. However, undergraduate students in Chinese universities have demonstrated only moderately high levels of learning engagement in the context of blended learning, which may be enhanced through the introduction of knowledge graph. This study aimed to examine whether students' learning engagement could be enhanced through knowledge graph-based blended learning. The findings are intended to provide a basis for implementing knowledge graph-based blended learning in future medical education.

Methods A quasi-experimental design was employed, involving a control group ($n=36$) of medical students from the 2020 cohort who received traditional blended learning and an experimental group ($n=50$) from the 2021 cohort. Knowledge graph-based blended learning was implemented in a core clinical medicine course, *Diagnostics*. The course was developed and implemented using the ADDIE (Analysis, Design, Development, Implementation, and Evaluation) instructional design model. Post-intervention, course grades and levels of learning engagement were compared between the two groups, and the learning satisfaction of the experimental group was analyzed.

Results The results indicated that the experimental group achieved significantly higher scores in the learning absorption dimension ($P=0.035$) and overall course grades. There was no statistically significant difference between the two groups in the total scores of learning engagement, dedication, and vigor ($P>0.05$). The average online learning time for students in the experimental group was 43.81 h, which demonstrated a positive correlation with their level of learning engagement ($r=0.843$, $P<0.001$). The overall learning satisfaction was high (94.69 ± 12.15) during the knowledge graph-based blended learning process.

Conclusion The application of knowledge graph-based blended learning in the Diagnostics course facilitates the improvement of medical students' learning absorption, academic performance, and teaching effectiveness.

†Bugang Li and Wenhui Hu contributed equally to this work.

*Correspondence:
Wei Chen
cweixy3@126.com
Yufei Li
liyufei666@163.com

Full list of author information is available at the end of the article

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Keywords Learning engagement, Knowledge graph, Blended learning, Diagnostics, ADDIE instructional design model

Introduction

Medical knowledge is inherently complex and fragmented, characterized by abstract mechanisms and varying degrees of complexity, presenting significant challenges for students to build coherent knowledge structures and develop critical thinking skills. Learning engagement is a persistent, positive, and fulfilling psychological state associated with learning, research, and employment, characterized by learning absorption, learning dedication, and learning vigor as three core dimensions. Learning engagement stimulates students' optimism, resilience, sense of meaning, and creativity [1]. Traditional pedagogical approaches often result in passive and superficial learning experiences, which are insufficient to foster the motivation and skills necessary for inquisitive exploration, thereby hindering the cultivation of higher-order and innovative thinking abilities. Research has shown that medical students' learning engagement significantly influences their academic performance [2, 3]. Another study demonstrated that high levels of learning engagement enhance students' satisfaction with their learning experiences [4]. Therefore, learning engagement plays a vital role in medical students' learning. Blended learning combines traditional face-to-face instruction with online learning modalities, providing a student-centered and more adaptable educational framework [5, 6]. A survey [7] revealed that Chinese university students demonstrated only moderate levels of learning engagement in online learning contexts. Specifically, cognitive engagement in online learning was moderate, emotional engagement was relatively high, while the behavioral engagement was comparatively low. Thus, there is considerable room for improvement in the level of learning engagement in blended learning.

To address this challenge and enhance academic performance, knowledge graph was used in our study. Google introduced the knowledge graph in 2012, providing a formalized framework for describing entities and relationships within a knowledge domain, thereby revolutionizing the field of knowledge representation [8, 9]. Knowledge graph visualizes the logical interconnections between entities and facilitates knowledge reasoning [10]. The educational knowledge graph can effectively integrate fragmented knowledge, assisting learners in visualizing and organizing the relationships among knowledge points within a cohesive system [11]. Concurrently, knowledge graph can motivate students to engage in active thinking, assist them in autonomously constructing knowledge systems and enhance the proactive nature of their pre-class learning [12]. However, the efficacy of knowledge graph in improving medical students' learning engagement remains unclear. Moreover, the effect of knowledge graph-based blended learning in Diagnostics course has not been reported. Blended learning based on the ADDIE model have enhanced medical staff's theoretical and practical operational aspects [13]. Therefore, this study aimed to take the Diagnostics course as an example, employing the ADDIE instructional design model to investigate the impact of knowledge graph-based blended learning on medical students' learning engagement.

Methods

Participants

From September 2023 to February 2024, with informed consent from the students, we utilized a cluster sampling method to select the 2021 undergraduate class majoring in preventive medicine at Hunan Normal University Health Science Center as the experimental group ($n = 50$). The experimental group received instruction through using a knowledge graph-based blended learning, while the 2020 undergraduate class majoring in preventive medicine served as the control group ($n = 36$), receiving instruction through traditional lecture-based methods.

Study design

This study employed a quasi-experimental research. *Diagnostics* was selected as the course to implement knowledge graph-based blended learning in this study. *Diagnostics* is a core clinical medicine course that plays a pivotal role in bridging basic medical education and clinical medicine education [14].

Teaching methods

Experimental group

The blended learning method for *Diagnostics* was supported by the knowledge graph and guided by the ADDIE instructional design model. The specific teaching process is illustrated in Fig. 1.

Control group

Traditional blended learning methods were employed. The traditional blended learning process included: (1) pre-class online self-study, where students independently watched instructional videos and reviewed course materials via the Chaoxing Learning Platform on computers or mobile devices and completed pre-class quizzes; (2) in-class face-to-face teaching, where instructors explained the content using a combination of blackboard and multimedia tools, guiding students to reflect and gradually resolve common and challenging problems

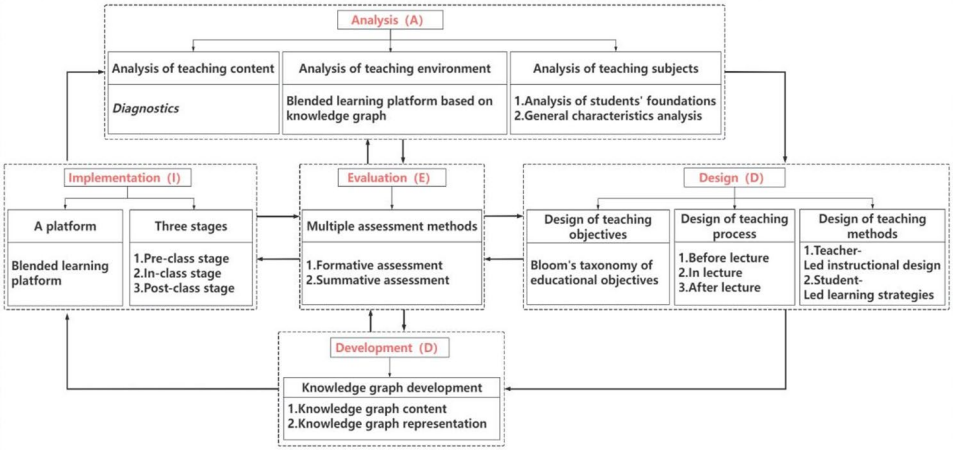


Fig. 1 Flowchart of the blended learning design for *Diagnostics*, based on the knowledge graph and the ADDIE instructional design model

from the pre-class assignments; (3) post-class online consolidation, during which students completed assessments on the Chaoxing Learning Platform and independently consulted relevant literature to keep abreast of the latest research advances.

Analysis ① Analysis of Teaching Content: The textbook used was *Diagnostics* (9th edition), published by People's Medical Publishing House. ② Analysis of Teaching Environment: The teaching environment in this study primarily relied on a blended learning platform based on the knowledge graph. This platform was built on the Chaoxing. Learning Platform, where teachers constructed the knowledge graph of the Diagnostics course, enabling students to independently and repeatedly study the course content through the knowledge graph. ③ Analysis of Teaching Subjects: Analysis of Students' Foundations: Examination of students' existing knowledge and skill mastery; Analysis of General Characteristics: Consideration of factors such as age, gender, and educational background.

Design ① Design of Teaching Objectives: Based on an "output-oriented" talent training system, the teaching objectives were aligned with graduation requirements and specific indicators, forming a one-to-one support relationship. Following Bloom's taxonomy of educational objectives, the objectives were further refined to focus on cultivating of higher-order abilities. ② Design of Teaching Process: Details are provided in the specific implementation process. ③ Design of Teaching Methods: This includes teacher-led instructional design and student-led learning strategies. The teacher-led instructional design primarily employed lecture-based teaching and question-and-answer methods, while student-led learning strategies focus on autonomous learning and practical exercises.

Development Based on the previous analysis and design, teachers constructed a knowledge graph of the Diagnostics course on the blended learning platform (Fig. 2).

The knowledge graph reorganized diagnostic knowledge according to its logical structure. The ontology of the course content integrated the textbook's table of contents with its inherent logic, categorizing it into five levels of knowledge granularity: course knowledge, knowledge systems, knowledge units, knowledge sets, and knowledge points. Ultimately, eight primary themes (knowledge systems) were derived, further refined into 77 secondary themes (knowledge units), 342 tertiary themes (knowledge sets), and 1,089 quaternary themes (knowledge points), totaling 4,262 knowledge granules (Fig. 3).

Within the teaching context of this study, the knowledge graph is essentially a structured "knowledge map" with links to corresponding learning resources. Unlike traditional linear syllabi or tables of contents, its core function lies in visually representing the logical relationships among all knowledge elements (termed "entities") in Diagnostics, thereby integrating fragmented diagnostic knowledge into an interconnected network system. In this structure, solid arrows indicate parent-child relationships, dashed arrows represent pre-requisite and post-requisite relationships, and dotted lines denote associative relationships (Fig. 4). Following Bloom's taxonomy of cognitive educational objectives, a tiered set of teaching goals was crafted for each class, advancing from lower to higher cognitive levels and embracing a student-centered pedagogy. The objectives include "remembering", "understanding", "applying", "analyzing", "synthesizing", and "evaluating". Additionally, based on the teaching objectives and competency goals of *Diagnostics*, emphasis was placed on identifying "key points", annotating teaching data as "difficult points", outlining examination topics as "examination points" for qualification examination, and incorporating ideological

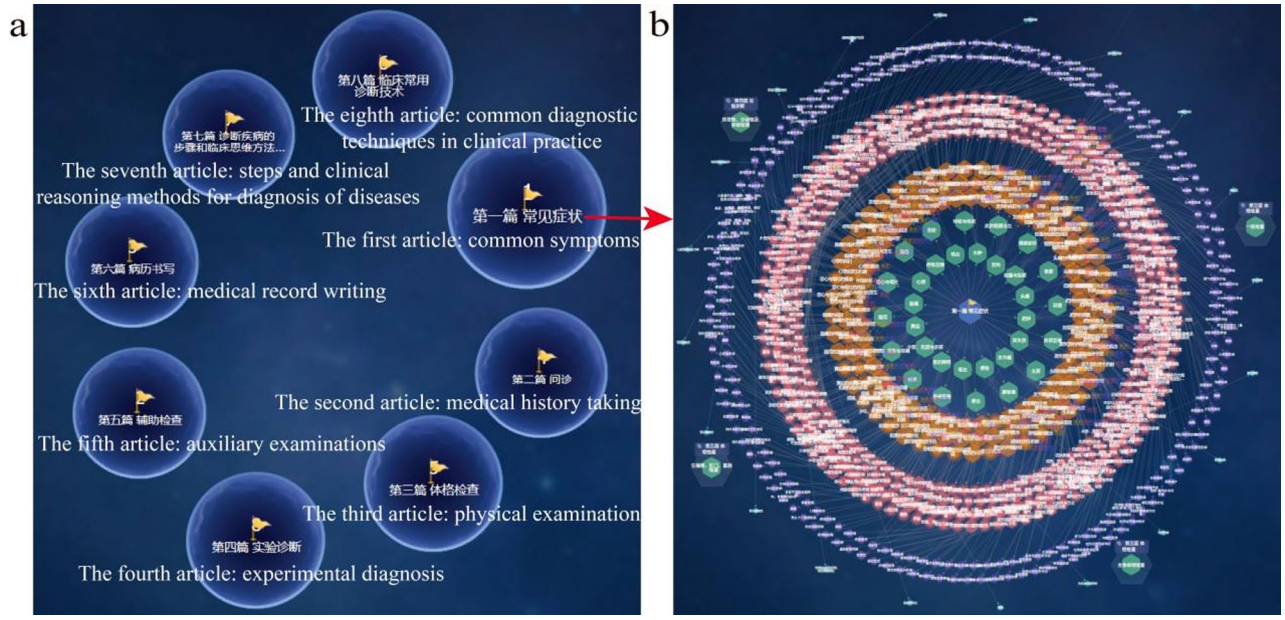


Fig. 2 Overview of the knowledge graph of *Diagnostics*. (a). Schematic diagram of the eight knowledge systems of *Diagnostics*; (b). Schematic diagram of the knowledge graph of common symptoms in *Diagnostics*, from the innermost to the outermost layer, they are: knowledge system, knowledge units, knowledge sets, knowledge points, and knowledge granules

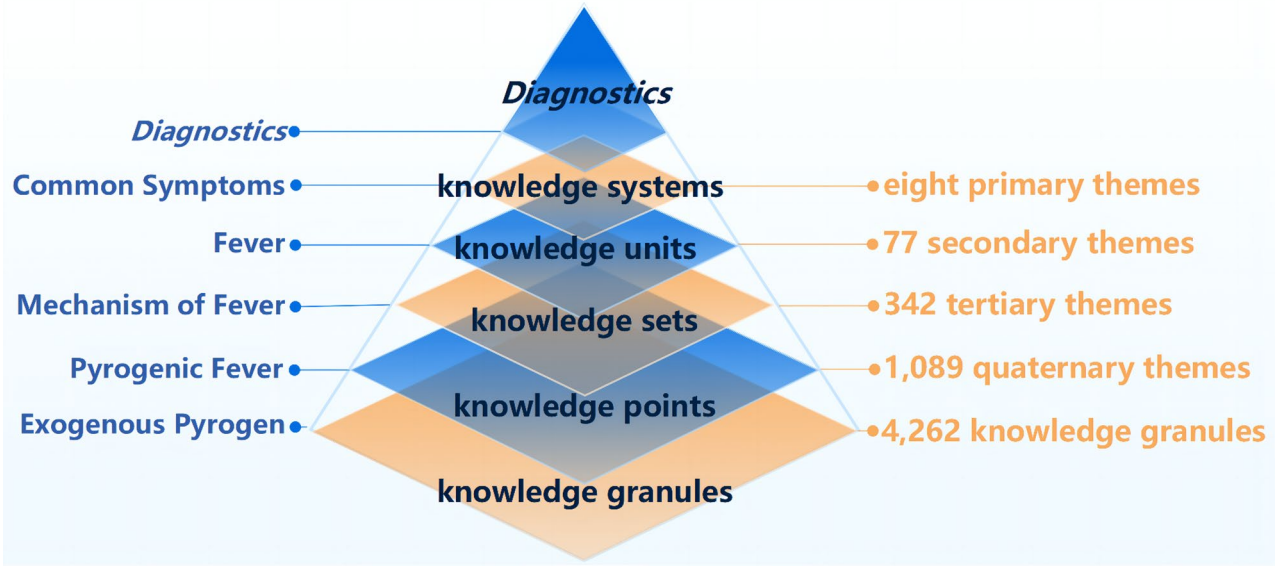


Fig. 3 The logical structure of the diagnostic knowledge graph, using "Fever" as an example

and political requirements as "ideological and political knowledge points". This course includes 4,262 knowledge granules. The main interface of knowledge graph supports both precise and fuzzy keyword-based retrieval, facilitating the efficient location of knowledge. It also integrates knowledge with educational resources, including textbooks, outlines, presentations, teaching plans, course videos, and more. The Diagnostics course encompasses 217 chapters, accompanied by 4518 questions, 369 videos, 84 documents, 57 supplementary materials, and 1 audio resource. Clicking on a specific knowledge unit, such as "Fever," will immediately display all related learning resources (e.g., instructional videos, audio clips, documents, and test questions). It will also show the subsets of "Fever," their prerequisite and post-requisite relationships, as well as associated knowledge elements (Fig. 4). Students can directly click to learn. The backend platform enables educators to monitor, students' engagement with knowledge points in real time.

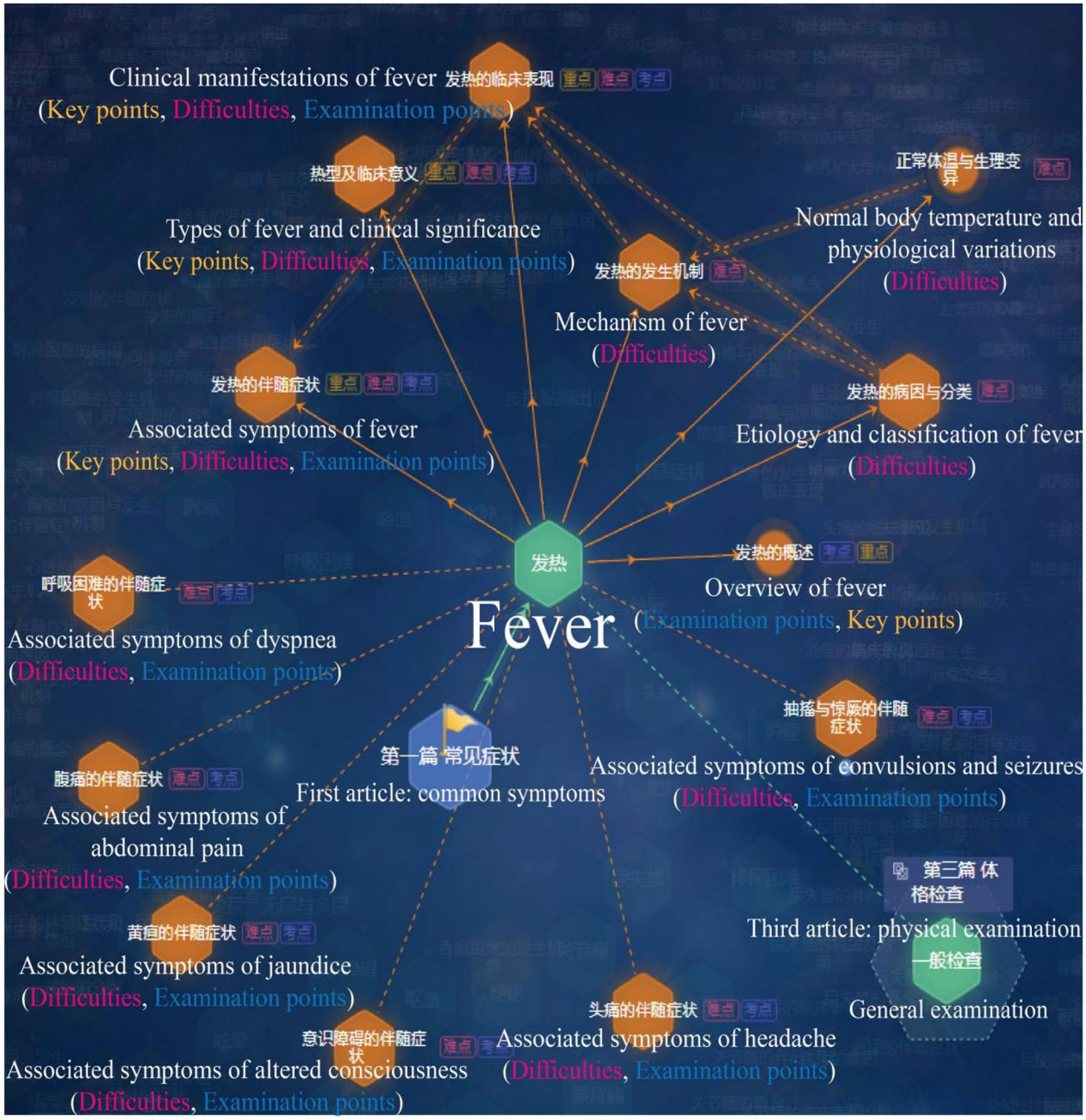


Fig. 4 Diagram of the entities and knowledge relationships of the "Fever" knowledge unit in *Diagnostics*

Implementation Based on the analysis, design, and development content, a process for implementing blended learning was formulated (Fig. 5). The detailed procedure of the blended learning process based on the knowledge graph is summarized in Table 1.

Evaluation Within the ADDIE instructional design framework, the evaluation phase incorporates both formative and summative assessments. Throughout the teaching process, teachers employed a suite of evaluative strategies to gauge students' comprehension of the course and the extent to which the instructional objectives were achieved, including pre-class assessments, pre-class discussions, case analyses, in-class quizzes, and post-class assessments. Post-curricular analysis included a comparative assessment of academic performance and engagement levels between the control and experimental cohorts. Additionally, the experimental group' satisfaction with the blended learning modality was collected and evaluated.

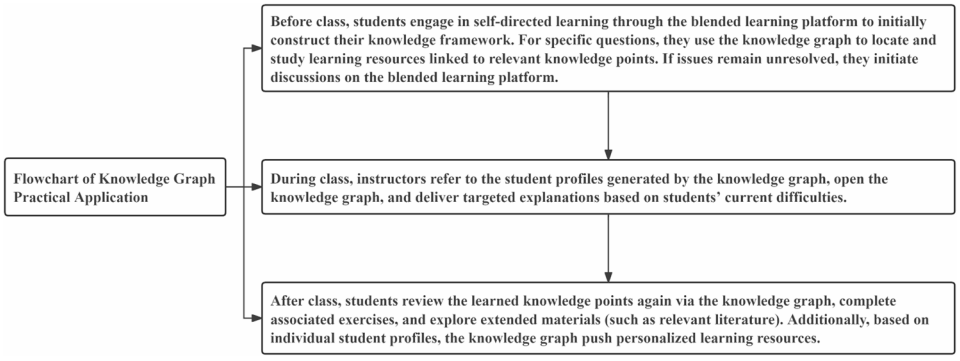


Fig. 5 Flowchart of knowledge graph practical application

Table 1 The implementation process of blended learning based on the knowledge graph

The teaching segment	Implementation process
Prior to class	λ Students logged into the blended learning platform to engage in online learning of relevant knowledge units, guided by the knowledge graph constructed by the teacher. λ Participants completed pre-chapter assessments using the provided resources to evaluate their comprehension and mastery of the knowledge units. λ Based on pre-assessments and feedback, teachers utilized the knowledge graph to clarify students' learning trajectories and pedagogical emphases.
In class	λ Teachers assisted students in constructing their knowledge systems by aligning various teaching objective levels with the weakness identified points through pre-class knowledge graph analysis. λ In the final ten minutes of the class, teachers prompted students to consult the knowledge graph to locate and respond to in-class quizzes targeting specific weakness previously identified within the knowledge units.
After class	λ Students engaged in summarization exercises and to undertook post-class evaluations for the knowledge units addressed within the knowledge graph. λ Learning resources were tailored and disseminated to individual students based on their unique profiles and learning requirements. λ Students were tasked with studying pertinent scholarly literature corresponding to the knowledge units delineated by the knowledge graph.

Measurement instruments

Course grades

Course grades were determined by 60% to the final examination and 40% to continuous assessment, culminating in a total score of 100 points. The final examination, conducted as a closed-book test after the curriculum, encompassed a variety of question types, including multiple-choice, true/false, definitions, short-answer, and case analysis questions, with a total score of 100 points. Continuous assessment scores, reflecting attendance and task completion, were directly extracted from the blended learning platform, with a total score of 100 points. The two components of the scores were evaluated using standardized criteria established by the teachers.

Utrecht work engagement scale-student (UWES-S)

The study utilized the Chinese version of the UWES-S, initially developed by Schaufeli et al. [15]. Subsequently, it was revised by Fang et al. [1] to assess the comparative levels of learning engagement between cohorts. The scale comprised three dimensions-learning absorption, learning dedication and learning vigor-with 17 items: six items in the learning absorption dimension, five in the learning dedication dimension, and six in the learning vigor dimension. Participants were required to select options that accurately reflected their perceptions of the statements in each section. Responses ranged from "strongly disagree" to "strongly agree" on a seven-point Likert scale (i.e., 1 = "strongly disagree, " 4 = "neither agree nor disagree, " and 7 = "strongly agree"), with higher scores indicating greater levels of learning engagement. The Cronbach's α coefficient of the scale was 0.95, and the internal consistency coefficients for each dimension ranged from 0.82 to 0.95, indicating robust reliability and validity.

Satisfaction questionnaire

The satisfaction of the experimental group was assessed by the satisfaction questionnaire developed by Zheng [16], drawing upon prior scholarly work [17, 18]. The questionnaire comprised four dimensions with 21 items: four in the platform dimension, five in the course dimension, six in the interaction dimension, and six in the resources dimension. Participants were required to select options that accurately reflected their perceptions of the statements in each section. Responses were scored on a five-point Likert scale, ranging from 1 ("strongly dissatisfied") to 5 ("strongly satisfied"). The percentage-based grading system was used with higher scores indicating greater satisfaction levels. The Cronbach's α coefficient of the scale was 0.988, and the Kaiser-Meyer-Olkin (*KMO*) value was 0.978, indicating good reliability and validity.

Table 2 Comparison of learning engagement levels between two groups (M (P25, P75)/mean ± SD)

	Learning absorption	Learning dedication	Learning vigor	Learning engagement
Experimental group ($n=50$)	31.00 (24.75, 35.25)	25.00 (22.75, 29.25)	25.00 (20.75, 30.00)	79.14 ± 20.77
Control group ($n=36$)	29.00 (25.00, 30.00)	24.00 (20.00, 27.00)	23.00 (19.25, 29.75)	75.86 ± 14.92
Z/*t*-value	−2.110	−1.553	−0.658	0.808
p-value	0.035	0.120	0.511	0.421

Data analysis

Two individuals input Excel to establish a database and input the data. Data analysis was performed using SPSS Statistics 26.0 software. Descriptive analyses included means, standard deviations (SD), medians, and interquartile ranges. Normality tests were conducted for continuous variables, and those following a normal distribution were presented as mean ± SD. Variables not following a normal distribution were represented using the median and interquartile range. Comparison of course grades and learning engagement levels between the two groups were conducted using independent samples *t*-tests or non-parametric Mann-Whitney *U* tests. Predetermined significance was set at $P<0.05$.

Results

Baseline characteristics

No statistically significant differences were observed between the two groups in terms of age, gender, educational level, or clinical basic course scores ($P>0.05$).

Course scores of experimental group and control group

The course scores of the experimental and control groups were compared at the end of the survey. The experimental group achieved a mean course score of 79.96 ± 6.77 points, while the control group scored 76.78 ± 7.78 points. The course score of the experimental group was higher than that of the control group ($P=0.047$).

Learning engagement levels of experimental group and control group

Data from the blended learning platform indicated that the experimental group students spent an average of 43.81 h on online learning, with the primary learning period occurring between 16:00 and 24:00. Pearson correlation analysis showed a positive correlation between online learning time and learning engagement level ($r=0.843$, $P<0.001$).

Subsequently, the total scores of learning engagement and the scores of each dimension were compared between the experimental and control groups at the end of the course. The total learning engagement score did not differ significantly between the two groups ($P>0.05$) (Table 2). Among the three dimensions of learning engagement, only the score on the learning absorption dimension reached statistical significance ($P=0.035$) (Table 2).

Furthermore, within the experimental group, students in the top quartile of diagnostic scores exhibited significantly higher levels of learning engagement compared to those in the bottom quartile ($P=0.004$).

Table 3 Overall learning satisfaction score during the experimental group blended learning process (mean ± SD, points)

Norm	Scores
Platform	17.69 ± 2.76
Course	22.39 ± 3.44
Interaction	27.39 ± 3.54
Resources	27.22 ± 3.25
Overall learning satisfaction	94.69 ± 12.15

Overall satisfaction of the experimental group

The overall learning satisfaction score and its constituent indicators were evaluated to assess the experimental group's satisfaction with the knowledge graph-based blended learning approach. The total score for overall learning satisfaction was 94.69 ± 12.15, indicating a relatively high level of student satisfaction (Table 3).

Discussion

In the current era of digital education, knowledge graph is increasingly used in the educational field. Their application in medical education demonstrates unique and novel value. The complexity and fragmentation of medical knowledge, along with the need to develop higher-order clinical thinking, has made knowledge graph more than just simple teaching aids. Instead, they have transformed into a core solution addressing the key challenges in medical education. Their novelty is mainly reflected in three aspects: First, they restructure the presentation logic of medical knowledge through semantic network relationships, directly fostering students' clinical relational thinking. Second, through fine-grained knowledge tracing, they enable scalable and personalized instruction for the first time in large courses such as *Diagnostics*. Third, they possess the potential to seamlessly integrate basic and clinical disciplines, providing a technical blueprint for future reforms in integrated medical curricula.

Knowledge graph-based blended learning is conducive to improving medical students' learning absorption

As a powerful tool, the knowledge graph establishes connections between entities and captures intrinsic relationships within massive information, presenting them

in a systematic and visual manner [19]. This directly impacts students' cognitive processes by reducing the cognitive load required for knowledge acquisition and integration. It enables students to perceive connections between knowledge points more clearly [20], thereby immersing them more easily in understanding the structure and logic of the knowledge itself, which constitutes the core connotation of learning absorption. Knowledge graph-based blended learning differs significantly from traditional blended learning. Before class, students independently and repeatedly study using the knowledge graph within the blended learning platform. The structured preview and on-demand access reduce the aimlessness of preparation, enhance sense of purpose and efficiency, making them easier to enter a focused state and laying the foundation for subsequent learning. During class, instructors utilize the knowledge profiles generated by the knowledge graph to quickly identify knowledge weaknesses at both group and individual levels, enabling more precise emphasis on key difficulties and relational points during lectures and guidance [21]. This "targeted" teaching approach better captures students' attention and reduces distractions caused by content that is either too simple or too fragmented. After class, personalized learning resources and knowledge point-based targeted exercises are pushed according to each student's knowledge profile. This enables efficient and focused review and consolidation. Such strengthening exercises, which concentrate on specific weaknesses, require a high degree of absorption. Furthermore, in the knowledge graph-based blended learning environment, students' online learning time is concentrated primarily in the afternoon and evening, with an average of 43.81 h. A positive correlation was observed between online learning duration and level of learning engagement. This duration exceeded in the average reported by Morris et al. [22]. By leveraging collaborative educational technology and facilitation, students can actively engaged in learning, enhance intrinsic motivation, improve learning outcomes, and foster engagement in educational programs [23]. With increased participation, students invest more in learning, making their learning motivations and objectives more explicit, and devote more time and energy to their studies [24].

However, this study did not observe significant improvement in the two dimensions of learning engagement—dedication and vigor—as a result of the knowledge graph intervention. Learning absorption primarily reflects cognitive investment and immersion during learning. Learning dedication, on the other hand, emphasizes emotional investment, sense of meaning, enthusiasm, and pride in learning. Learning vigor refers to the level of energy, mental resilience, and willingness to persist during learning [15]. The knowledge graph-based blended learning designed in this study primarily optimizes the organization of knowledge, access pathways, and instructional efficiency. Its direct impact focuses more on cognitive processes and information processing, while its effect on stimulating deeper learning enthusiasm or enhancing fundamental energy levels may be limited. Furthermore, the intervention was implemented over a single semester. The UWES-S is a well-established scale that measures relatively stable psychological traits. A short-term course intervention may be insufficient to significantly alter the perception of such stable traits at the group level, particularly in dimensions more related to affective and energetic states such as learning dedication and learning vigor [25]. Lastly, learning vigor is particularly susceptible to external factors such as course load, overall physical and mental health status, and daily routine regularity. Although this study controlled for baseline characteristics, it did not specifically measure or control for these variables, which may have a stronger influence on learning vigor. In the future, we will link learning resources within the knowledge graph entities that help cultivate medical students' sense of national commitment and medical professionalism, thereby further enhancing their learning dedication. At the same time, we will continue to optimize the presentation of the knowledge graph. For example, we have currently designed a new learning map. Taking the knowledge unit "Fever" as an example: the learning map first displays an "Overview of Fever." After completing this section, students can proceed to the next stage—"Mechanism of Fever"—progressing step by step in a structured manner. This checkpoint-based learning mode aims to stimulate students' interest and enhance their learning vigor.

Knowledge graph-based blended learning significantly improves medical students' academic performance while enhancing their learning absorption

Previous study has indicated that learning absorption has a significant positive effect on academic performance [1]. In this study, the post-intervention course scores of the experimental group were higher than those of the control group ($P < 0.05$), which confirms earlier research findings. Shi et al. [12] applied a knowledge graph-based enhanced blended learning approach to teach the *Paleozoic Invertebrate Zoology* course. The findings showed that students correctly retained over 60% of the learned knowledge without review. By the end of the course, most students were proficient in comprehensively summarizing the learned knowledge using the knowledge graph approach, and they gained a clearer understanding of the overall structure and logic of *Paleozoic Invertebrate Zoology*. A previous study demonstrated that the newly developed intelligent multilevel knowledge graph attracted the attention of online learners and improved

their learning outcomes [26]. These findings demonstrate that knowledge graphs facilitate the summarization and comprehension of knowledge points, promoting systematic learning, comprehensive knowledge mastery, and the development of logical thinking skills.

Personalized learning is a key application of knowledge graphs in education [27]. Each student possesses unique learning preferences, strengths, and areas of comprehension gaps. Knowledge graphs customize learning pathways based on individual progress and needs through semantic representation and intelligent analysis. This enables them to determine optimal learning sequences, recommend relevant resources, and adapt learning experiences to each student's pace and style [28]. Additonally, students can also use knowledge graph test results to identify and address deficiencies while exploring related and hidden knowledge points for further learning. These facilitate enhancing students' understanding, flexibly applying learning content, and continually improving learning outcomes [11]. Therefore, the improvement in students' academic performance is a natural outcome of enhanced learning absorption and the knowledge graph's role in promoting systematic understanding and application of knowledge.

Medical students are more satisfied with knowledge graph-based blended learning method

The overall satisfaction score for the blended learning process in the experimental group was 94.69 ± 12.15 points, indicating a high level of student satisfaction with this teaching model. Our findings are consistent with those reported in other studies, such as Sun et al. [29] constructed a knowledge graph-based system to recommend Chinese vocabulary learning resources, addressing the vocabulary learning difficulties of beginner Chinese language learners while enhancing their learning efficiency and vocabulary application abilities. Survey results indicated high satisfaction among learners with the recommendation system. Jiang et al. [30] developed a dynamic knowledge graph model for deep learning, which significantly improved students' perceptual attention, confidence, and satisfaction during task completion. Weng et al. [31] developed a teaching system based on crowdsourced knowledge graphs, and research indicated that this system addressed the shortcomings of traditional adaptive learning systems and was widely accepted by both teachers and students.

The reasons for students' satisfaction with knowledge graph-based blended learning may be attributed to the following factors: (1) The structural advantages of the knowledge graph enhance the learning experience. Integrating knowledge graphs into course design and planning provides a data-driven, learner-centered approach that enhance the quality, relevance, and effectiveness of teaching outcomes [32, 33]. For instance, Yang et al. [34] constructed a knowledge graph to help water resources students understand the relationships among courses, knowledge units, and terms. It aimed to provide structured representations of water resources education data for efficient learning and knowledge retrieval. (2) Knowledge graph visualization has significant capabilities in enhancing engagement, understanding, and exploration of complex educational content [28]. Tang et al. [35] developed a knowledge graph to visually depict the course content system, integrate online learning resources, establish correspondences between knowledge points and resources, and facilitate students' rapid discover and understanding of interrelationships among knowledge points. (3) The functional design of the blended learning platform improves operational satisfaction. The knowledge graph supports precise or fuzzy keyword searches, enabling quick access to related knowledge points and increasing efficiency compared to traditional platforms. The system monitors the completion status of knowledge nodes, allowing students to clearly track their progress and enhancing their sense of self-efficacy. Overall, students' high satisfaction with knowledge graph-based blended learning essentially stems from the close alignment between the instructional design and the principles of medical learning.

Limitations and strengths

The study developed a knowledge graph for Diagnostics and achieved a closed-loop integration of the ADDIE model with knowledge graph-based blended learning. Both objective and subjective evaluations were conducted to assess its effectiveness. However, several limitations should also be noted to guide future improvements.

The student cohorts in 2020 and 2021 were relatively small and differed in size. The use of different academic years for the experimental and control groups may introduce a potential cohort effect. Additionally, the lack of random sampling could lead to selection bias. These factors may affect the interpretation of the results. Although baseline equivalence was reported for some metrics, this does not preclude differences in other unmeasured variables. The initial levels of learning engagement were not assessed in either the experimental or control groups prior to the teaching intervention. In addition, teaching satisfaction was evaluated only in the experimental group, which may introduce bias into the findings.

In conclusion, future efforts should focus on optimizing the presentation of the knowledge graph, incorporating larger and more diverse student populations, adopting randomized sampling methods, and implementing knowledge graph-based blended learning across various courses and institutions to further explore its potential benefits in medical education.

Conclusion

This study demonstrated that applying the knowledge graph-based blended learning in the Diagnostics course improved teaching effectiveness for medical students, particularly in terms of learning absorption and outcomes. Furthermore, medical students reported high levels of satisfaction with this teaching model. This study represents a valuable advancement in blended learning, offering new insights into the development of student-centered teaching models. These benefits reflect their recognition and support for this teaching method, which largely explains the improvement in students' absorption and academic performance. It is hoped that this teaching method can be applied to other medical courses in the future.

Abbreviations

ADDIE	Analysis/Design/Development/Implementation/Evaluation
UWES-S	Utrecht work engagement scale-student
KMO	Kaiser-Meyer-Olkin
SD	Standard deviations
SPSS Statistics 26.0	Statistical Product and Service Solutions Version 26.0

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12909-025-08428-5.

Supplementary Material 1.

Supplementary Material 2.

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Authors' contributions

Y.F.L., W.C., B.G.L. and W.H.H. conceived the original idea and wrote the proposal. Y.F.L., W.C., B.G.L., W.H.H., C.Y.L., L.L., C.Y.W. and Z.K.P. designed and applied the study. B.G.L., W.H.H., A.Y.D. and Q.L. organized the data collection and analyzed the data. All the authors contributed to revising the draft of the manuscript, and all the authors read and approved the final draft of the manuscript.

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Data availability

Complete survey data used/analyzed in this study are available from the corresponding author on request.

Declarations

Ethics approval and consent to participate

The study obtained all participants' informed consent before participating and ethical approval from the Institutional Review Committee of Hunan Normal University School of Medicine before commencing the study (Ref no 2023479, dated 1 st November 2023). We certify that the study was performed in accordance with the 1964 Declaration of Helsinki and later amendments.

Consent for publication

All authors consented to the publication of the study.

Competing interests

The authors declare no competing interests.

Author details

[1]Hunan Normal University Health Science Center, 371 Tongzipo Road, Changsha, Hunan 410013, China
[2]Kiang Wu Nursing College of Macau, Macao, China
[3]The Affiliated Changsha Hospital of Xiangya School of Medicine, Central South University, Changsha, Hunan, China
[4]The Third Xiangya Hospital of Central South University, Changsha, Hunan, China

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